SUNSET ON THE BERNÁBEU

The LaLonde Arc

How the field lost its faith, and how it found it again

Scott Cunningham

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In 1986, a 32-year-old in Princeton asked the wrong question

> *If economists swap the experimental controls for a non-experimental comparison group, do their methods still recover the true treatment effect?*
>
> — **Robert LaLonde, *AER*, 1986**

A practical question. A heretical implication. He answered: ***no****.*

Why this mattered

Most causal claims in economics are observational. If matching the wrong way produces the wrong answer, then **every observational claim is under suspicion**.

The question that broke the field

Experimental truth → Same treated units → Different controls (CPS) → New estimate

If the new estimate = the truth, observational methods work.

If not, the field has a problem.

The National Supported Work Demonstration (1975–1979)

- Run by the **Manpower Demonstration Research Corporation**
- Random assignment of long-term unemployed to subsidized jobs
- Treated: 9–12 months of guaranteed work + counseling
- **Eligible applicants randomized**: 297 treated, 425 controls
- Outcome: 1978 annual earnings (`re78`)
- Dehejia–Wahba (1999) restricted sample: **185 treated, 260 controls**

Why it was credible

Not a textbook RCT. A real public program. Real frictions. **Random assignment is the only thing rescuing the answer from selection bias.**

The data fits in 11 columns and 445 rows

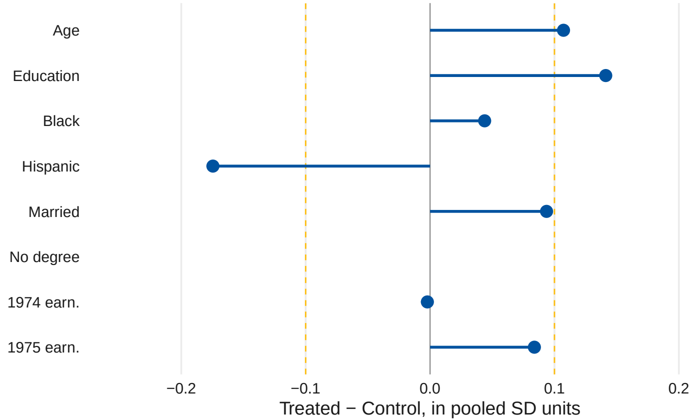
```r
library(haven); library(dplyr)
nsw <- read_dta("nsw_mixtape.dta")

#   data_id          treat   age   educ  black  hisp  marr  nodegree    re74    re75
    re78
#   Dehejia-Wahba      1      37    11      1     0     1      1          0       0
    9930
#   Dehejia-Wahba      1      22     9      0     1     0      1          0       0
    3596
#   Dehejia-Wahba      1      30    12      1     0     0      0          0       0
    24909

exp_att <- mean(nsw$re78[nsw$treat==1]) -
           mean(nsw$re78[nsw$treat==0])
#> $1,794
```

One line of R. One number. The number we're going to compare everything else to.

$1,794

THE EXPERIMENTAL ATT

Random assignment did its job



Standardized differences in every covariate within ± 0.1. The two groups are interchangeable on observables. We will leverage that fact, and then we will lose it.

Who was Robert LaLonde?

- **Princeton**, Ph.D. 1985 (advisor: Orley Ashenfelter)
- Assistant professor at Princeton, 1985–1991
- Later: Michigan State, then Chicago Harris (2000–2017)
- **Died in 2018 at age 60** — before he saw the full arc of what his *AER* paper would set in motion

He was a labor economist who became famous for testing labor econometrics. The provocation was his career's defining act.

The 1986 paper

Evaluating the Econometric Evaluations of Training Programs with Experimental Data

American Economic Review, March 1986

$\sim$11,000 citations (as of 2025)

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The Swap

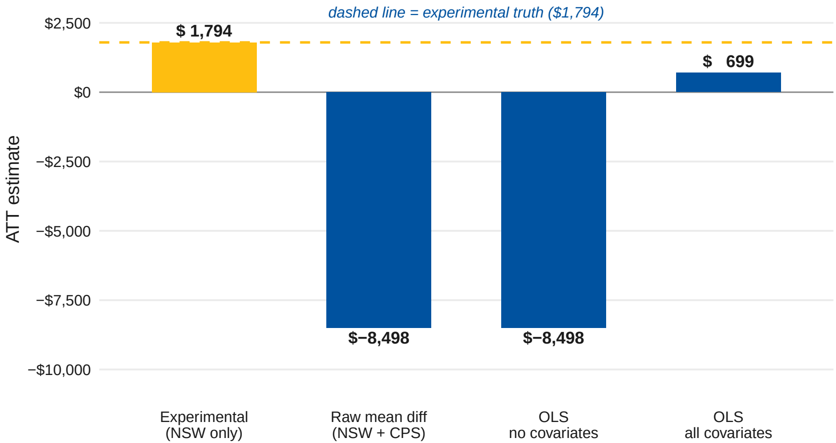
The move: keep the treated, replace the controls

EXPERIMENT: NSW treated (185) → NSW controls (260) → ATT: $1,794

SWAP: NSW treated (185) → CPS controls (15,992) → ATT: ?

Same treated units. Different counterfactual. Standard econometric methods.

The naïve methods collapse



Every standard estimator misses by thousands of dollars. OLS-with-controls comes closest — and is still off by $1,095.

$-\$8{,}498$

RAW BIAS FROM THE CPS SWAP

The skeptic's reading was the obvious one

> *If a state-of-the-art econometric toolkit, given a clean RCT to shadow, cannot recover the experimental answer — what business do we have using these methods on observational data alone?*
>
> — **the field, 1986**

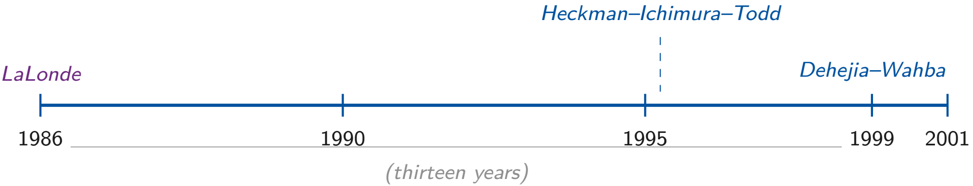
What cannot be fixed

Functional form. Controls. Significance stars. Bigger samples.

What must be re-thought

Identification. Selection. The assumption behind OLS.

Then nothing happened for thirteen years



The toolkit existed: Heckman (1979), Rosenbaum–Rubin (1983), matching. ***Off the shelf, none of it closed the gap.***

The 1986 toolkit could adjust — but it could not diagnose

Heckman (1979) · **Rosenbaum–Rubin (1983)** · **standard matching**

The gap

Every method on the shelf in 1986 *adjusted* for observed X. None of them told you **whether the adjustment had worked**. That is what LaLonde demanded — and what the field couldn't deliver for the next decade.

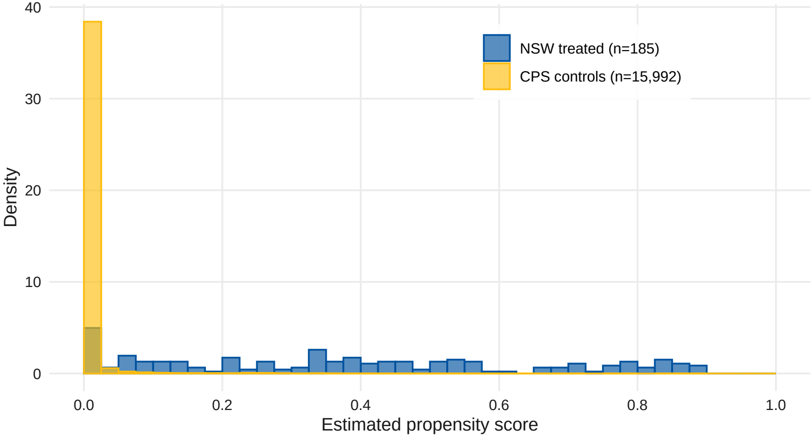
The first crack: Dehejia & Wahba (1999, JASA)

- Re-ran LaLonde, but with **propensity score matching**
- Subsampled NSW to the 185-treated/260-control overlap window
- PSM brought the ATT **back to roughly \$1,600–\$1,900**
- First published demonstration that an observational method could recover the experimental answer

But matching alone is fragile

- Sensitive to the PS specification
- Sensitive to which units are in the support window
- Smith & Todd (2005): change covariates, change the answer

The treated and controls live in different worlds



93% of CPS controls have $\hat{p} < 0.01$. ***This is the geometry that breaks matching*** *— and the geometry the next two methods will navigate.*

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Heckman–Ichimura–Todd (1997)

Who were Heckman, Ichimura, and Todd?

Heckman	Ichimura	Todd
Chicago, the Nobel laureate (2000) *Selection theory, training-program evaluation, the field's senior figure*	Pittsburgh, then Tokyo *Semiparametric estimation, single-index models*	Penn, then Chicago Booth *Heckman student. The empirical engine of the trio.*

Review of Economic Studies, October 1997: *a 50-page paper that reframed matching as a diagnostic problem, not just an estimator.*

The HIT diagnostic: total bias has three sources

$$B = E[Y^0 \mid D{=}1] - E[Y^0 \mid D{=}0]$$

B_1**: non-overlap**

Treated outside control support

B_2**: density**

Different distribution of X

B_3**: selection**

Selection on **unobservables**

Matching can fix B_1 and B_2. Only conditional DiD can fix B_3.

The key insight: conditional parallel trends is weaker

- **Matching** requires $E[Y^0 \mid X, D{=}1] = E[Y^0 \mid X, D{=}0]$
  Levels *of the untreated outcome match conditional on* X*.*
- **Conditional DiD** requires only
  $E[Y_t^0 - Y_{t'}^0 \mid X, D{=}1] = E[Y_t^0 - Y_{t'}^0 \mid X, D{=}0]$
  Changes *match conditional on* X*. Level differences are allowed.*

What HIT showed empirically

On the JTPA data with multiple comparison groups: matching's assumption (**A**-**3**, **A**-**4**) was **rejected at** $p < 0.001$. The DiD assumption (**A**-**5**) was **not rejected**.

The estimator: local-linear kernel matching on ΔY

1. Fit $\hat{p}(X) = \Pr(D{=}1 \mid X)$ via logit
2. Restrict to common support: $\hat{p} \in [\hat{p}_{\min}, \hat{p}_{\max}]$
3. For each treated i: estimate $\widehat{E}[\Delta Y \mid \hat{p}, D{=}0]$ using **Fan's (1992) local linear regression**
4. Average $\Delta Y_i - \widehat{E}[\Delta Y_i \mid \hat{p}_i, D{=}0]$ over treated units

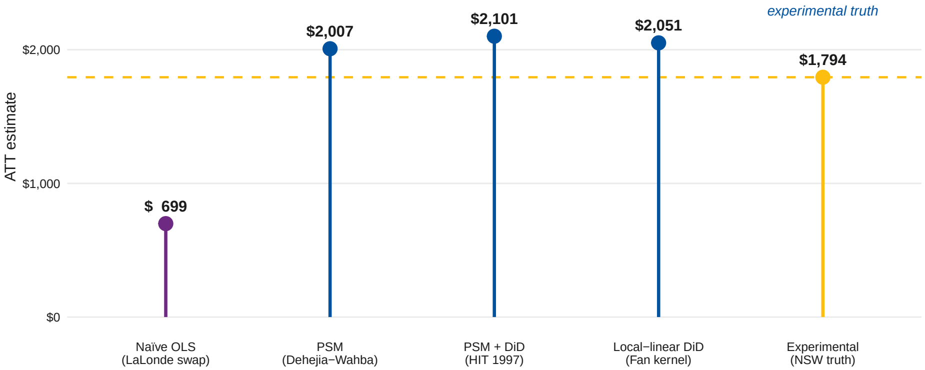
Why each step matters

- Step 2 fixes B_1 (non-overlap)
- Step 3 fixes B_2 (density weighting) via kernel re-weighting
- Step 4 fixes B_3 if A-5 holds (conditional parallel trends)

Twelve lines of R get you to $2,051

```
ps_mod <- glm(treat ~ age + I(age^2) + educ + black + hisp +
              marr + nodegree + re74 + re75 + I(re74^2) + I(re75^2),
              data=cps, family=binomial)
cps$ps <- predict(ps_mod, type="response")
cs <- subset(cps, ps>=min(ps[treat==1]) & ps<=max(ps[treat==1]))
trt  <- subset(cs, treat==1);  trt$dy  <- trt$re78  - trt$re75
ctrl <- subset(cs, treat==0);  ctrl$dy <- ctrl$re78 - ctrl$re75
h <- 0.9 * min(sd(ctrl$ps), IQR(ctrl$ps)/1.34) * length(ctrl$ps)^(-1/5)
trt$dy0 <- sapply(trt$ps, local_linear, x=ctrl$ps, y=ctrl$dy, h=h)
hit_att <- mean(trt$dy - trt$dy0)
#> $2,051
```

The first road home



$2,051

HIT RECOVERS THE EXPERIMENTAL NEIGHBORHOOD

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Abadie (2005)

Who is Alberto Abadie?

- MIT Ph.D. 1999 (advisor: Joshua Angrist)
- Harvard Kennedy School 2000–2018, then MIT economics
- **Inventor of the synthetic control method** (with Gardeazabal, 2003)
- Co-developer of the design-based standard-error literature (Abadie–Athey–Imbens–Wooldridge, 2020)

A Spaniard. From the Basque country. Codechella Madrid would not exist without him.

The 2005 paper

Semiparametric Difference-in-Differences Estimators

Review of Economic Studies, January 2005

∼3,500 citations

Re-weight instead of match

- HIT estimates **one function for the treated** and **one for the controls**, then differences them
- Abadie estimates **one function only** — the propensity score $p(X)$ — and uses it as a **re-weighting kernel**

The headline identity

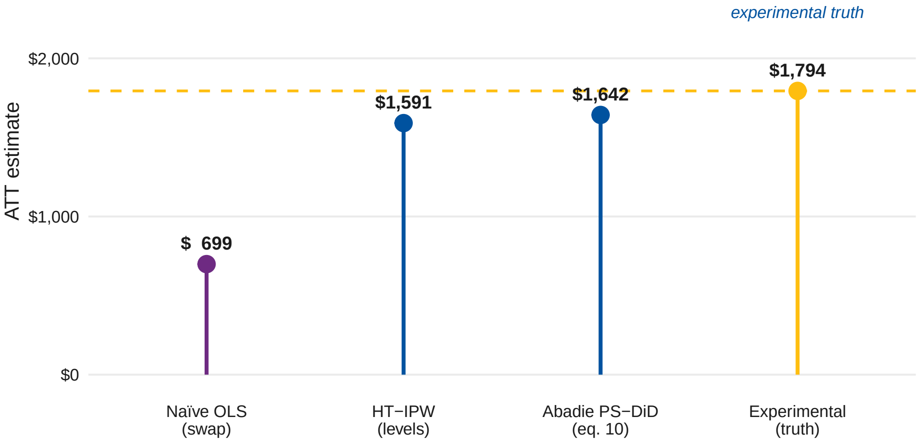
$$\text{ATT} \;=\; \frac{1}{\Pr(D{=}1)}\, E\left[\frac{D - p(X)}{1 - p(X)} \cdot \big(Y_1 - Y_0\big)\right]$$

One propensity score. One weighted mean of temporal differences. Done.

Five lines of R get you to $1,642

```r
ps_mod <- glm(treat ~ age + I(age^2) + educ + black + hisp +
              marr + nodegree + re74 + re75 + I(re74^2) + I(re75^2),
              data=cps, family=binomial)
cps$ps <- predict(ps_mod, type="response")
cs <- subset(cps, ps>0.01 & ps<0.99)
cs$dY <- cs$re78 - cs$re75
abadie_att <- mean( ((cs$treat - cs$ps)/(1 - cs$ps)) * cs$dY ) / mean(cs$treat
    )
#> $1,642
```

The second road home



$1,642

ABADIE ALSO RECOVERS THE EXPERIMENTAL NEIGHBORHOOD

Two different machines, same destination

Method	ATT	Gap vs. truth
LaLonde's swap (the crisis)		
OLS, all covariates	\$699	\$-1,095
The two roads home (the resolution)		
Heckman–Ichimura–Todd (1997)	\$2,051	\$257
Abadie (2005) PS-weighted DiD	\$1,642	\$-152
Experimental benchmark	**\$1,794**	—

Match-then-difference and **re-weight-and-difference** converge on the same neighborhood as the experimental answer.

When two independent roads built a decade apart arrive at the same place, the answer is in.

The roads converge on a neighborhood, not a point

- HIT delivers **$2,051**; Abadie delivers **$1,642**. The two roads disagree with each other by **$409**.
- That gap is the price of specification choice: the propensity-score functional form, the trimming rule, the kernel bandwidth.
- **Smith & Todd (2005, JoE)**: change the covariates, change the answer. Neither method is a free lunch.

What is actually robust

Not the third significant digit. What is robust is the **direction** ($1,500–$2,100, not negative) and the **order of magnitude** (thousands, not tens of thousands). That is enough to declare LaLonde's challenge answered.

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Why Then?

Why HIT in 1997 — not 1987, not 1995?

- **Rosenbaum–Rubin (1983)** gave the field the propensity score. *But it took a decade to be absorbed.*
- **Fan (1992, 1993)** made local-linear kernel regression respectable. *HIT needed this — ordinary kernel matching has bias that LL eliminates.*
- **Card–Krueger (1994)** revived DiD as a serious identification tool. *Suddenly, the idea of differencing across time was in the air again.*
- **Eleven years** since LaLonde had passed. The unsolved challenge had become a **career-defining target** for an ambitious empirical team.

The Heckman lab in the 1990s

- Heckman won the Nobel in 2000 for selection models, but his Chicago lab in the 1990s was a **dissertation factory**
- Dissertations that came out of it: Petra Todd, Hidehiko Ichimura, Jeffrey Smith
- The lab had been chewing on the LaLonde puzzle for years
- The 1997 paper was followed by **Heckman–Ichimura–Smith–Todd (1998, ReStud)** — the companion piece with the full theory

The lesson

Big methodological breakthroughs come from **labs**, not lone genius. Heckman gave the architecture; Ichimura and Todd built it.

Why Abadie in 2005 — not 1995, not 2015?

- **Hirano–Imbens–Ridder (2003)** showed PS weighting is semiparametrically efficient. *Suddenly re-weighting had a respectability that matching alone lacked.*
- **Bertrand–Duflo–Mullainathan (2004)** broke DiD inference and forced the field to rebuild the standard error machine. *Abadie's PS-DiD slotted into the rebuild.*
- **Abadie–Gardeazabal (2003)** — synthetic control. Abadie himself had spent two years thinking about **weights as the solution to identification**, not matches.
- The 2005 paper was the natural synthesis: take HIT's conditional DiD assumption, replace the matching step with re-weighting, prove the asymptotics.

A-5 is strictly weaker than A-4 — and that is the whole story

A-4 (matching)

$$E[Y^0 \mid X, D{=}1] \;=\; E[Y^0 \mid X, D{=}0]$$

***levels* match**

implies →

A-5 (conditional DiD)

$$E[\Delta Y^0 \mid X, D{=}1] \;=\; E[\Delta Y^0 \mid X, D{=}0]$$

***changes* match; level shifts allowed**

*Selection on time-invariant unobservables is allowed under **A-5**. The level of Y^0 can differ between treated and controls — only the time path must match. That weakening is what HIT proved you needed, and what Abadie 2005 then leveraged.*

The Abadie weight ρ_0

Define $\rho_0 = (D - p(X))/(p(X)\,(1 - p(X)))$. Then under **A-2** and **A-5**,

$$E\left[\rho_0 \cdot (Y_1 - Y_0) \,\middle|\, X\right] \;=\; E[Y^1 - Y^0 \mid X, D{=}1]$$

Reading the weight

- Treated $(D = 1)$: $\rho_0 = 1/p(X)$ — upweight rare-treated cells
- Controls $(D = 0)$: $\rho_0 = -1/(1 - p(X))$ — downweight common controls
- $E[\rho_0 \mid X] = 0$ — weights sum to zero in expectation

And so — back to the three numbers.

Three numbers, one truth

$1,794

$2,051

is roughly equal to

$1,642

Experimental $\approx$ Heckman–Ichimura–Todd (1997) $\approx$ Abadie (2005)

THE LALONDE ARC

When two roads built a decade apart arrive at the same $1,794, the field has its answer.